section 6.4 of these Guidelines. These measurements should be in accordance with chapter 5 of NTC 2008.
6.2.2 Paragraph 5.9.8.1 of the NTC 2008 requires engine conditions to be measured at each mode point, for an engine system . This equally applies in the case of an engine fitted with SCR. Additionally, exhaust gas temperature at the intended inlet of the SCR chamber should be determined and recorded in the test report as required by section 5.10 of NTC 2008.
6.3 Test procedures for SCR chambers
6.3.1 General
6.3.1.1 The SCR chamber for validation testing may be either a full-scale SCR chamber or a scaled version. A SCR chamber should demonstrate the reduction in NO x concentrations

---CHUNK---

SOURCE: MEPC.399(83).pdf
6.3 Test procedures for SCR chambers
6.3.1 General
6.3.1.1 The SCR chamber for validation testing may be either a full-scale SCR chamber or a scaled version. A SCR chamber should demonstrate the reduction in NO x concentrations
(ppm) expected in exhaust gas measured in section 6.2 of these Guidelines. Therefore, the NOx reduction rate of the SCR chamber should be determined for each individual mode point. Where undertaken on a scaled version of the SCR chamber the scaling process should be validated to the satisfaction of the Administration.

6.3.1.2 The scaling process is to correspond with the modelling tool of paragraph 6.1.1.2 of these Guidelines, and take into account geometrical reference conditions, and chemical NO x conversion models, and other parameters which have an influence on the NOx conversion rate in the modelling tool. If the scaling process could not be validated satisfactorily by theoretical

---CHUNK---

SOURCE: MEPC.399(83).pdf
conversion models, and other parameters which have an influence on the NOx conversion rate in the modelling tool. If the scaling process could not be validated satisfactorily by theoretical
analysis or calculations taking into consideration the complex conditions in the SCR chamber, such as uniformity of gas speed and reductant, a combined engine and SCR system validation test in accordance with Scheme A should be undertaken.
6.3.1.3 The modelling tool of paragraph 6.1.1.2 of these Guidelines is acceptable for use in other engine groups which operate within the same defined boundary conditions.
6.3.2 Test conditions at each mode point
Exhaust gas, catalyst, reductant and an injection system should satisfy the following conditions at each mode point:
.1 Exhaust gas flow
Exhaust gas flow rate for the test should be scaled accordingly to account for the dimension of the catalyst model.
.2 Exhaust gas component
Exhaust gas for the test should either be diesel engine exhaust gas or simulated gas.
Where diesel exhaust gas is used, it should correspond, in terms of concentrations, to the exhaust gas in section 6.2 of these Guidelines, in

---CHUNK---

SOURCE: MEPC.399(83).pdf
Exhaust gas for the test should either be diesel engine exhaust gas or simulated gas.
Where diesel exhaust gas is used, it should correspond, in terms of concentrations, to the exhaust gas in section 6.2 of these Guidelines, in terms of NOx, O2, CO2, H2O and SO2 (±5% of the required concentration for each emission species).
Where simulated gas is used, it should correspond, in terms of concentrations, to the exhaust gas in section 6.2 of these Guidelines, in terms of NO, NO2, O2, CO2, H2O and SO2 (±5% of the required concentration for each emission species) balance N2.
An exemption for one or more of the above -mentioned gas species' concentration requirements may be allowed subject to a demonstration test showing that the gas or gases do not affect the NO x reduction rate by more than 2%.
.3 Exhaust gas temperature
The temperature of exhaust gas used for the test should correspond to the temperatures obtained from testing in section 6.2 of these Guidelines, ensuring that the SCR chamber is activated at every load point , other than as provided for by 3.1.4 of the NTC 2008, and that no ammonia bisulphate formation, or reductant destruction, takes place.

---CHUNK---

SOURCE: MEPC.399(83).pdf
ensuring that the SCR chamber is activated at every load point , other than as provided for by 3.1.4 of the NTC 2008, and that no ammonia bisulphate formation, or reductant destruction, takes place.
.4 Catalyst blocks and AV, SV value
The catalyst blocks used in the test should be representative of the catalyst blocks to be used in the SCR chamber in service. AV, SV or LV value should, in the case of full scale tests, be within -5% or above of the required value as obtained in testing from section 6.2 of these Guidelines. In the case of scaled tests it should correspond to the above.
RESOLUTION MEPC.399(83) (adopted on 11 April 2025)
2025 GUIDELINES ON SELECTIVE CATALYTIC REDUCTION (SCR) SYSTEMSMEPC 83/17/Add.1
Annex 3, page 14
I:\MEPC\MEPC 83-17-Add.1.docx
.5 Reductant
The reductant concentration on the surface of the tested catalyst should be representative of the reductant concentration on the surface of the catalyst during actual engine operation. Ammonia gas may be used as a reductant for the SCR chamber test, provided that it results in an equivalent concentration on the catalyst surface.
6.3.3 Stability for measurement

---CHUNK---

SOURCE: MEPC.399(83).pdf
during actual engine operation. Ammonia gas may be used as a reductant for the SCR chamber test, provided that it results in an equivalent concentration on the catalyst surface.
6.3.3 Stability for measurement

All measurements should be recorded after they have stabilized.
6.3.4 List of data to be derived from the model
6.3.4.1 Operating data which is to be given in the Technical File should be derived from the modelling process or otherwise justified.
6.3.4.2 Exhaust gas analysers should be in accordance with appendix III and appendix IV of NTC 2008 or otherwise to the satisfaction of the Administration.
6.3.5 Test report for SCR chamber
Data recorded under paragraph 6.3.1.1 of these Guidelines should be recorded in the test report as required by section 5.10 of NTC 2008.
6.4 Calculation of the specific emission
6.4.1 The NOx emission value of the engine system fitted with SCR should be calculated as follows:
( )( )
( )

=
=
=
=

= n i
i F
F
n i
i mgasi
x
i
ii
W P
W q
1 i
1 100100gas
Where:
i NOx reduction rate (%) derived in accordance with section 6.3 of these Guidelines;
imgasq
= Mass flow of NO x gas measured in accordance with

---CHUNK---

SOURCE: MEPC.399(83).pdf
( )( )
( )

=
=
=
=

= n i
i F
F

n i
i mgasi
x
i
ii
W P
W q
1 i
1 100100gas
Where:
i NOx reduction rate (%) derived in accordance with section 6.3 of these Guidelines;
imgasq
= Mass flow of NO x gas measured in accordance with section 6.2 of these Guidelines;
iFW
= Weighting factor;
iP
= Measured power at individual mode points in accordance with section 6.2 of these Guidelines.
The weighting factors and number of modes (n) used in the above calculation shall be according to the provisions of section 3.2 of the NTC 2008.